SandBAR: A Band-Asymmetric Architecture

Supplementary Material

S1. Fold Precision

The bias fold of Eq. (8) is algebraically exact. With the two band branches contributing zero, the routed fusion reduces to the constant $\mathbf{b}_{\text{out}}$, and adding that constant to the affine bias of the preceding LayerNorm reproduces it term for term. The deployed graph is therefore the same function as the graph it replaces, expressed with $4{,}498{,}208$ fewer parameters.

To confirm that the identity survives floating-point execution, we compare the deployed folded graph against the composite graph on every one of the 490 test images. The two agree to 7.4×10^{-4} in the worst pixel of the worst image and to 1.5×10^{-5} on average, which is 19% and 0.4% of a single 8-bit quantisation step. In PSNR the largest disagreement on any image is 0.009 dB and the mean is 0.0003 dB. Folding therefore leaves the prediction intact at the precision at which images are stored and metrics are computed, demonstrating the benefit of choosing a scaffold whose removal is an algebraic operation rather than an approximation.

S2. Timing Procedure

Latency and peak memory in Table 2 are measured on one RTX 5090 at batch 1. Each graph is built once, moved to the device, and run for 10 warm-up passes before 100 timed forward passes under `torch.no_grad`, with `torch.cuda.synchronize` before and after the timed region; the reported figure is the mean of those passes. All graphs are timed in a single process and in the order in which they appear in the table, so the measurements share one allocator state and one clock. Peak memory is read from `torch.cuda.max_memory_allocated` after the timed region. FLOPs are counted with `torch.utils.flop_counter`, which counts two floating-point operations per multiply–accumulate.

S3. Training Configuration

All models are implemented in PyTorch and trained on a single GPU with Adam (β_1=0.9, β_2=0.999, ε=10^{-8}) at an initial learning rate of 10^{-4}, with three warm-up epochs followed by cosine annealing to 10^{-6} over the remaining budget, batch size 4 on random 256×256 crops with random flips and rotations, and gradient-norm clipping at 0.01. Headline models are trained for 650 epochs and the controlled architectural ablations for 100, each annealing over its own budget. The architecture uses $\text{dim} = 32$, blocks $(2, 4, 4, 6)$, heads $[1, 2, 4, 8]$, FFN expansion 2.66, bias-free convolutions and WITHBIAS LayerNorm throughout, and activation recomputation to hold the full block depth at batch 4 on one card.

Table S1. **Architectural ablation under additional metrics.** Three-domain mean PSNR, SSIM and LPIPS for the architectural variants from Table 5, evaluated over the same 490-image split.

Variant	PSNR↑	SSIM↑	LPIPS↓
(a) SandBAR full	**26.721**	**0.9098**	**0.1195**
(b) fixed Gaussian	25.734	0.9005	0.1296
(c) symmetric ops	26.286	0.9073	0.1256
(d) no LF guidance	26.280	0.9046	0.1306

S4. Ablation Under Additional Metrics

Table 5 reports the architectural ablation in three-domain mean PSNR. To establish that each design decision earns its place on structural and perceptual grounds as well, we score the same four models on SSIM and LPIPS over the same 490-image split, with the results shown in Table S1.

Every variant is worse than the full model on all three measures. Replacing the content-adaptive splitter with a fixed Gaussian costs 3.69% in PSNR, 1.02% in SSIM and 8.45% in LPIPS; making the two band operators symmetric costs 1.63%, 0.27% and 5.10%; removing the low-frequency guidance path costs 1.65%, 0.57% and 9.29%. The perceptual penalty is consistently the largest of the three, exceeding the PSNR penalty by more than a factor of two in every case, demonstrating the benefit of band-asymmetric repair for the qualities a human observer reads first. These results confirm that each of SandBAR's three design decisions contributes across fidelity, structure and perceptual similarity alike rather than to a single measure.

S5. Occupying Both Slots

The sandwich block offers two positions, and Table 4 shows a band module doing more in the rear one. To establish that a single rear module is the right design rather than a partial one, we additionally train a variant that occupies both slots for every configuration of that table. The resulting three-way comparison is shown in Table S2.

Moving a band module from the front slot to the rear one raises the three-domain mean by 1.10%, 1.03% and 1.11% across the three configurations, a gain that holds for a second module and a second backbone alike. Occupying both slots never improves upon post-attention-only placement. For SPARK, the two configurations are effectively tied on both SandBAR (26.72 vs. 26.71 dB) and Restormer (26.44

Table S2. **Effect of occupying both feed-forward slots.** Three-domain mean PSNR under the controlled 100-epoch budget.

Configuration		Mean PSNR↑		
Module	Backbone	pre-attn.	post-attn.	both
SPARK	SandBAR	26.43	**26.72**	26.71
DFFN [S1]	SandBAR	26.25	**26.52**	26.24
SPARK	Restormer [S2]	26.15	**26.44**	26.44

vs. 26.44 dB), while for DFFN occupying both slots degrades performance from 26.52 to 26.24 dB. These results show that a single rear-slot module is sufficient, and that duplicating band-specific processing offers no additional benefit.

References

[S1] Lingshun Kong, Jiangxin Dong, Jianjun Ge, Mingqiang Li, and Jinshan Pan. Efficient frequency domain-based transformers for high-quality image deblurring. In *CVPR*, 2023.

[S2] Syed Waqas Zamir, Aditya Arora, Salman Khan, Munawar Hayat, Fahad Shahbaz Khan, and Ming-Hsuan Yang. Restormer: Efficient transformer for high-resolution image restoration. In *CVPR*, 2022.